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Sweet Spot Identification Study in Unconventional Gas Play in UAE

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Abstract

This paper builds upon the methodology for identifying natural fractures in carbonate reservoirs utilizing the mega onshore seismic data based on the borehole imager and advanced mud logging data combined with the seismic inversion results. The approach is applied to new areas of Diyab unconventional gas play that feature post-stack seismic inversion data, including porosity and acoustic impedance volumes, alongside newly drilled wells with long-term well tests. The objective is to evaluate the robustness, adaptability, and predictive power of the methodology across varying geological conditions and reservoir properties to be used as a sweet spot identification technique for the development drilling. The workflow combines seismic inversion attributes with structural interpretation and borehole image log calibration to delineate fracture-prone zones. The methodology was adapted to incorporate changes in seismic response across the field and adjusted using data from long well tests and new image logs. Key inversion parameters—such as low impedance and relatively high porosity anomalies—were cross-analyzed with different structural attributes. Hard data calibration based on advanced mud logging was applied to refine the fracture detection algorithm. A comparative analysis between the original calibration area and new zones helped to assess transferability of key fracture indicators and define limits of extrapolation. The extended application of the methodology to newly drilled areas demonstrated a high degree of consistency between predicted fracture zones, high porosity anomalies and dynamic well test results. In several cases, wells that intersected zones flagged as fractured by the seismic model exhibited enhanced productivity, stable well head pressure and relatively low decline rate. Notably, areas with anomalously low impedance proved to be reliable indicators of sweet spots, particularly when supported by fracture anomalies linked to the connected fracture networks. Refining the thresholds for the fracture attribute extraction improved the predictive accuracy of the model under different reservoir conditions. The integration of long-duration well test data played a crucial role in validating the spatial continuity and hydraulic connectivity of the interpreted fracture zones. The findings highlight the potential for seismic inversion data to enhance fracture prediction in unconventional carbonate fields and support more targeted infill drilling and reservoir management strategies. This work demonstrates the practical scalability of a previously published fracture identification workflow by extending it to new areas of a producing carbonate reservoir. By integrating seismic inversion products with dynamic test data and borehole image interpretation, the study provides a validated, transferable framework for field-wide fracture

characterization. The refined methodology contributes to better decision-making in reservoir development, especially during development drilling of the unconventional assets where new wells and updated seismic attributes offer opportunities for model calibration and optimization. The work aligns with ADIPEC's focus on maximizing recovery through innovation, data integration, and applied geoscience in complex subsurface environments.

Introduction

Identifying high-prospectivity zones in unconventional reservoirs is essential for optimizing hydrocarbon recovery and ensuring economic feasibility. These zones are characterized by elevated porosity and permeability, making them ideal targets for drilling and production operations. The synergy between seismic inversion and fracture analysis supports precision drilling strategies. This integrated methodology reduces the risk of drilling non-productive wells and enhances hydrocarbon extraction efficiency. It reflects the industry's shift toward quality-focused development, where each well is strategically placed to maximize output. Ultimately, identifying high-quality reservoir zones in unconventional fields demands a detailed understanding of petrophysical and geomechanical properties. The fusion of seismic inversion insights with fracture characterization forms a robust framework for targeting productive zones, leading to more efficient and economically viable hydrocarbon production.

This study is based on the previously developed approach (Mykhaylenko, 2024) of identification of natural fractures based on Mega Onshore seismic data covering all onshore part of the UAE calibrated by wellbore imager data, advanced mud gas, cuttings and drilling parameters. Current analysis expands the applicability of the fracture analysis over the new exploration & appraisal areas guided by the seismic inversion (porosity & AI), calibrated by the available long term production testing from the horizontal wells. Comprehensive analysis of the geological and completion parameters reveals potential drivers of enhanced productivity and identification of the sweet spots of the Diyab unconventional reservoir.

Geological Description of the Diyab Formation

The Diyab Formation, an Upper Jurassic carbonate succession (Oxfordian to Kimmeridgian), is the principal source rock for several prolific fields in the UAE and surrounding regions. It was deposited in an intra-shelf basin setting within the Arabian Plate, flanked by shallower carbonate platforms and subjected to periodic transgressions and regressions.

Stratigraphically, Diyab is subdivided into three key lithostratigraphic units: Tuwaiq Mountain, Hanifa, and Jubaila. The Tuwaiq Mountain and Hanifa members are of particular interest due to their relatively higher organic content and favorable rock fabric for gas generation and fracture development. These units consist of organic-rich laminated limestones and peloidal wackestones, often deposited under euxinic and anoxic conditions, especially in basin centers. (Vahrenkamp, 2015)

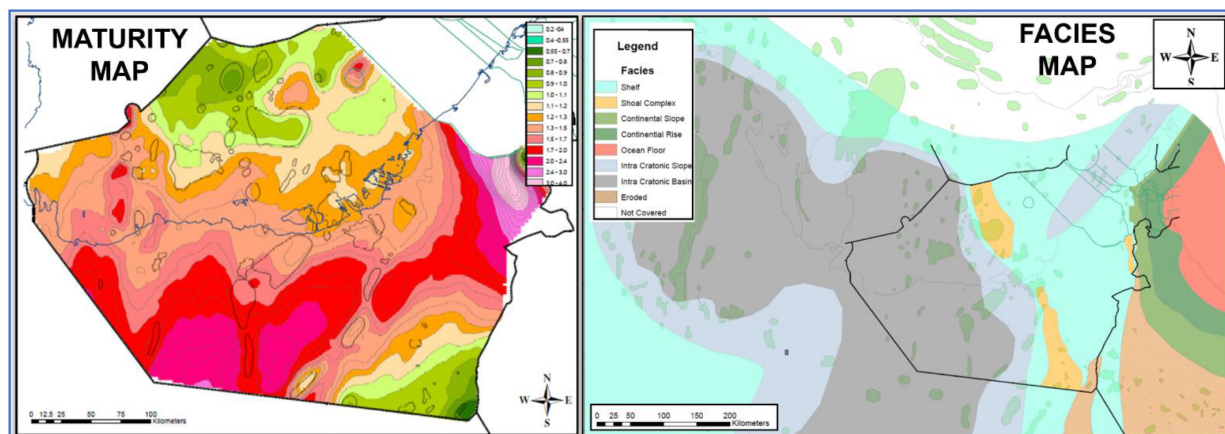


Figure 1—Diyab formation maturity and facies map

Core and seismic data reveal a well-developed anhydrite layer atop the Hanifa Formation, interpreted as a marker of a sea-level drop during late Oxfordian time. This subaqueous anhydrite bed—approximately 0.5 to 10 ft thick—was regionally deposited under restricted hypersaline conditions and has been correlated across the UAE and Qatar. It serves as a reliable time marker and may locally act as a mechanical barrier to hydraulic fracture propagation.

Regionally, the Diyab Formation exhibits significant lateral thickness and facies variability due to differential subsidence across the intrashelf basin. The Tuwaiq Mountain interval tends to thin northwestward and southwestward from the eastern basin margin toward more condensed basinal facies. Hanifa thickens slightly toward the eastern platform edge before transitioning into deeper, more organic-rich facies in the west.

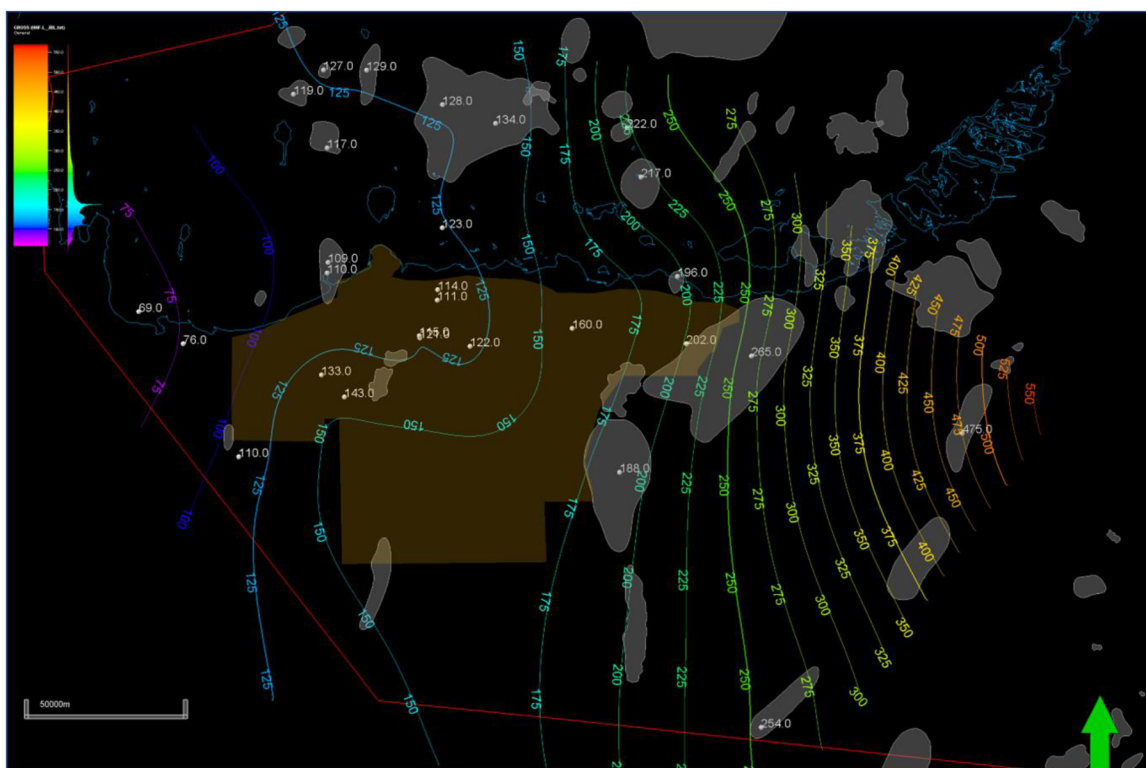


Figure 2—Thickness variation (ft) of the Hanifa interval based on well data.

Thermal maturity data from pyrolysis and vitrinite reflectance indicate that the Diyab Formation in the onshore UAE lies within the dry gas window. TOC-rich intervals show hydrogen indices below 50 mgHC/gTOC, with reflectance values $>1.2\%$ Ro, confirming over-maturity in most areas, especially in Hanifa and Tuwaiq Mountain members.

Formation evaluation of the Diyab

Petrophysical characterization of Diyab was carried out integrating the core data, high-resolution logs, advanced imaging (SEM, CT), and machine learning-based lithofacies classification workflows (Baig, M.Z., 2017, Alharthi, A., 2023) and expanded with the additional wells drilled in past two years. A heterogeneous rock analysis (HRA) framework enabled distinction of rock classes tied to TOC, porosity, and brittleness distributions, crucial for selecting hydraulic fracture targets. Gross thickness of the Diyab Formation ranges between 250 and 350 meters, with Tuwaiq Mountain and Hanifa members comprising roughly 100–150 meters each depending on location. Jubaila is relatively thicker but typically exhibits poorer reservoir quality. Porosity ranges from 2.7% to 8%, with Hanifa and Tuwaiq Mountain consistently displaying higher values than Jubaila. Maximum porosities were observed in wackestone facies with organic-rich laminations and intercrystalline porosity. TOC values exceed 4% in Hanifa and Tuwaiq Mountain, with Hanifa often reaching 6–8%. Jubaila, by contrast, averages $<1\%$ TOC. Permeability is extremely low, typical of unconventional systems, ranging from 10 to 100 nD. However, ambient permeability in cores shows a correlation with natural micro-fractures and organic-hosted pore networks, as confirmed by SEM and MICP analysis.

Water saturation in Hanifa and Tuwaiq mountain generally exceeds 20–30%, but localized desaturation is observed adjacent to naturally fractured zones. Brittleness, derived from elastic logs and supported by triaxial testing, correlates with carbonate content and appears highest in Tuwaiq Mountain's dense limestones.

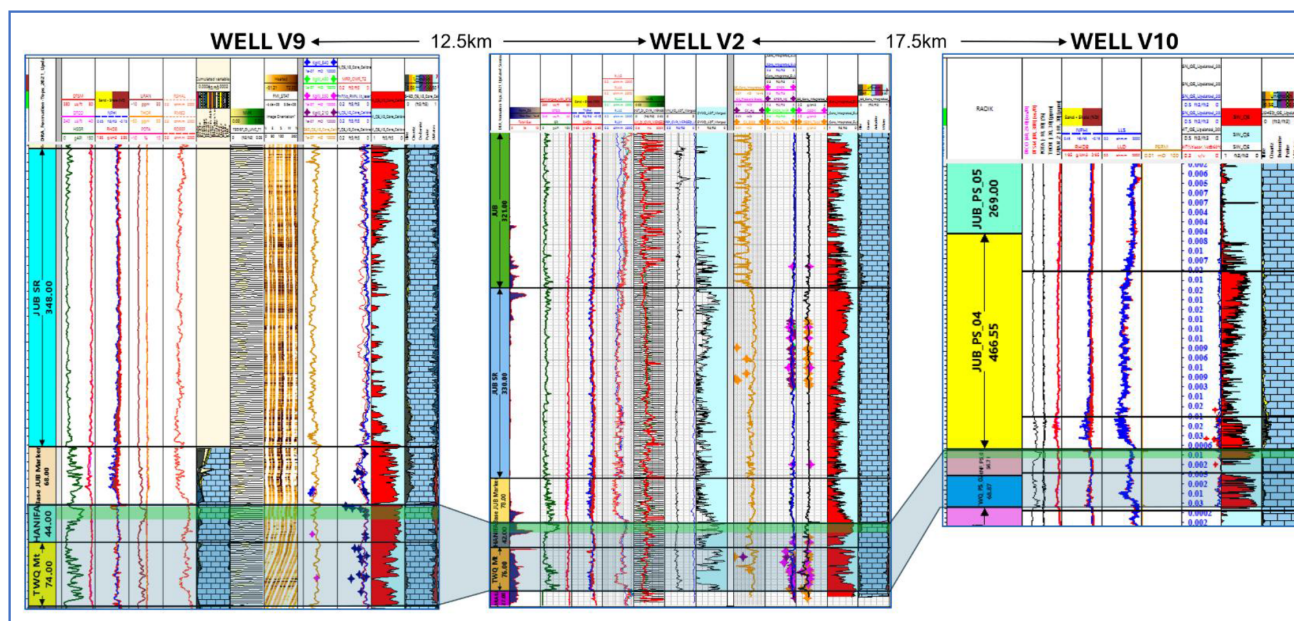


Figure 3—Petrophysical evaluation of the key Diyab wells.

Spatial trends in thickness, porosity and TOC derived from the seismic data correlate with the regional depositional model for Hanifa and Tuwaiq Mountain: the western basin center hosts more laminated, organic-rich but tighter facies; the eastern basin and mid-ramp feature more bioturbated, better connected porosity with slightly reduced TOC. This suggests a lateral shift from productive organic-rich units to

cleaner, less mature carbonates updip. Central part of the basin shows reduced reservoir thickness with increased porosity and condensed TOC sections. The presence of the anhydrite cap atop Hanifa also affects pore pressure buildup and compartmentalization. Core and log data suggest enhanced overpressure below this barrier, which is an important consideration for fracturing effectiveness.

Seismic reservoir characterization

Regional background. The tectonic evolution of Abu Dhabi's onshore is complex, involving multiple orogenic and extensional episodes. (Pascal Richard, 2017) These events have created a framework of structural domains, fault zones, and variable stress regimes that critically influence fracture development and reservoir productivity.

Three primary sets of faults dominate:

- NW–SE and NNW–SSE conjugate sets (transtensional origin, related to Alpine I)
- NE–SW and N–S basement-rooted faults (reactivated during compression)
- Cross faults developed during late fold growth (e.g., en echelon shears)

Seismic imaging shows many faults as composite zones of en echelon fractures rather than through-going planes, often stratigraphically limited and related to folding strain rather than basement tectonics.

The current stress field is transitional from strike-slip to reverse faulting depending on depth and location. In the Diyab Formation specifically, stress profiles suggest:

- Upper sections (e.g., Upper Jubaila) under reverse faulting regime ($S_v \leq S_{hmin} \leq S_{Hmax}$)
- Lower sections (Hanifa and Tuwaiq) dominated by strike-slip regime ($S_{hmin} \leq S_v \leq S_{Hmax}$).

The S_{Hmax} orientation is predominantly NE–SW to ENE–WSW, with slight variations depending on proximity to major structural elements or faults. Field data from the study area show S_{Hmax} orientation close to 320 degrees. The stress directions align with regional tectonic forces from Zagros compression and ophiolite emplacement.

Structural domains and present-day stress orientations significantly affect fracture propagation, stimulated rock volume (SRV), and proppant distribution. Observations from hydraulic fracture diagnostics show that:

- Fault intersections enhance stage efficiency if well oriented for shear slip.
- Zones with connected natural fractures and low closure stress favor SRV development.
- Areas with lateral heterogeneity in elastic moduli or stress anisotropy require customized completion strategies.

Study Area and Dataset Overview. The study area is situated in the onshore Ruwais region, west of Abu Dhabi, and is fully encompassed by a high-resolution Mega Onshore 3D seismic survey. The primary stratigraphic target is the Diyab Formation, encountered at an approximate depth exceeding 11,000+ feet. The well dataset comprises 15 wells, with borehole image logs serving as the principal source for quantifying fracture density. Supplementary data include drilling logs, well reports, cuttings analyses, and other relevant geological and petrophysical datasets. Pre-Stack Depth Migration (PSDM) in Time seismic volume was utilized for structural and fracture analysis, which offers enhanced seismic imaging quality at the target depth, characterized by superior signal-to-noise ratio and improved horizon continuity. This facilitates more reliable interpretation of sub-seismic scale discontinuities associated with natural fractures.

Authors faced challenges in fault and fracture analysis because there is no displacement of the seismic horizons at the Diyab target interval (11,000–13,000 ft). Major faults lie deeper in the basement, complicating extraction. The main task is to accurately identify major faults and locate the associated sub-seismic fracture zones. Geosteering cross sections of the 30 horizontal wells targeted Diyab didn't show any significant formation displacement, which can be interpreted as a fault, however, borehole imaging data indicates high density natural fracturing of the reservoir, considering high calcite content and brittleness. Regional seismic data can provide trend of small-scale formation discontinuities, which are related with the clusters of the fractures visible on imaging tools.

Fault identification workflow. Fault identification in 3D seismic data is a critical step in subsurface structural interpretation, reservoir characterization, and geohazard assessment. Modern workflows (Colleen, 2020) leverage a combination of seismic attributes to enhance fault visibility and automate detection. Authors utilized several structural attributes such as dip illumination, coherence, 3D curvature and others. Before attributes were calculated, preconditioning of seismic data was carried out, including noise attenuation and amplitude balancing, to improve attribute reliability. Dip illumination is then applied to highlight discontinuities in dip magnitude and azimuth. This attribute enhances fault visibility by emphasizing areas with the local change of the dip, often corresponding to fault planes or fracture zones. Coherence volume was then estimated for detecting fault edges and stratigraphic boundaries. Advanced coherence algorithms, such as semblance-based or gradient-based methods, can further refine fault detection by reducing false positives. Curvature was utilized to help delineate subtle flexures and associated fracture network. These attributes are especially valuable in identifying small-scale faults or fractures that may not be visible in conventional amplitude data.

Once attributes are computed, they are integrated into a fault enhancement volume (RGB blending) to isolate fault features for a more intuitive fault interpretation. Finally, fault extraction manual and automated algorithms were used to generate fault surfaces and fault sticks, which were utilized to build the structural model. This multi-attribute approach significantly improves fault detection accuracy and reduces interpretation time, enabling more robust structural models, which reduces the risk for the exploration and appraisal drilling.

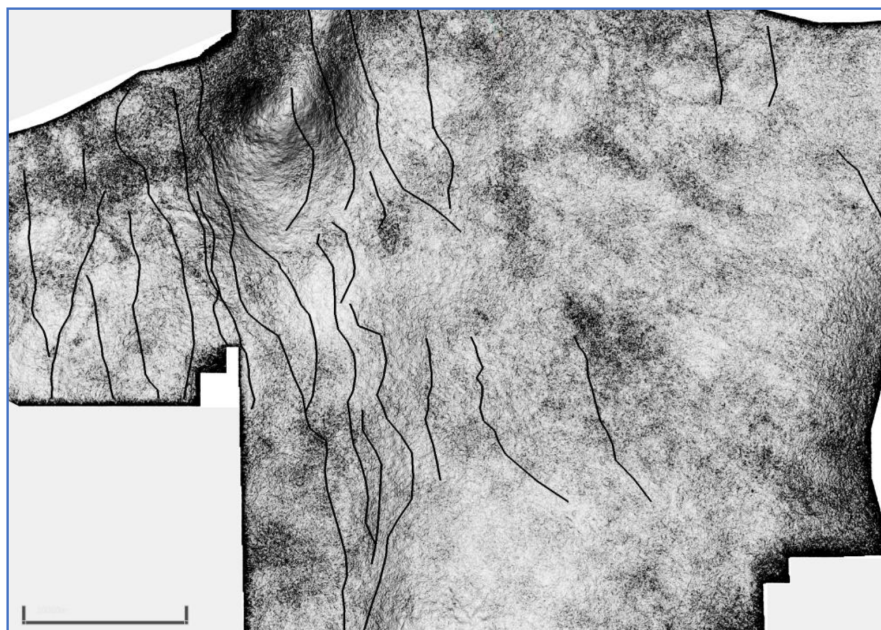


Figure 4—Fault model overlaid with Top Hanifa Structural slice (dip illumination)

Fracture identification workflow. Data Conditioning – Structural Smoothing. As detailed in (Mykhaylenko, 2024), Workflow #1 was adopted as the most robust methodology for predicting fracture networks in undrilled locations. This approach is calibrated using borehole image log and advanced mud logging data from existing wells. Details of the fracture extraction parameters are below:

Structural smoothing was applied to enhance the lateral continuity and coherence of seismic reflectors, particularly across faulted zones. This process improves the interpretability of fault geometries and their spatial relationships with surrounding stratigraphy. Alternatively, noise reduction algorithms were tested in different software showing comparable outputs.

Coherence attributes were computed to delineate structural discontinuities such as sub-seismic faults. Following the methodology of Bahorich (1995), low-coherence zones were interpreted as potential fracture zones due to disrupted seismic continuity. The coherence volume was generated using a $3 \times 3 \times 12$ filter window, optimized to retain structural discontinuities while suppressing random noise.

Ant-tracking was applied to the coherence volume to enhance the visibility of linear and planar discontinuities. Both passive and aggressive tracking parameters were used: Passive settings yielded fewer but more geologically reliable features. (Fig.5) Aggressive settings captured a broader spectrum of discontinuities, including subtle features potentially indicative of small-scale fracturing.

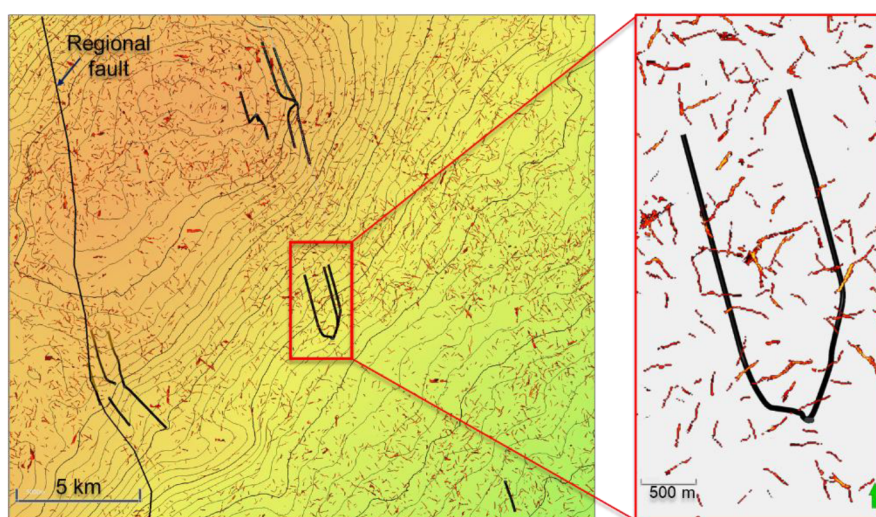


Figure 5—Top Hanifa Structural map overlaid with the Ant-tracks

Each individual well was evaluated in 2D and cross section (Fig. 6) views to analyze the spatial distribution and geological consistency of the extracted fractures. Wells with available imager and advanced log data were used as reference wells to analyze the natural fracture distribution in the rest of the wells. The quantity of the intersections of the fractures from seismic attributes with each lateral was summarized in the table 1.

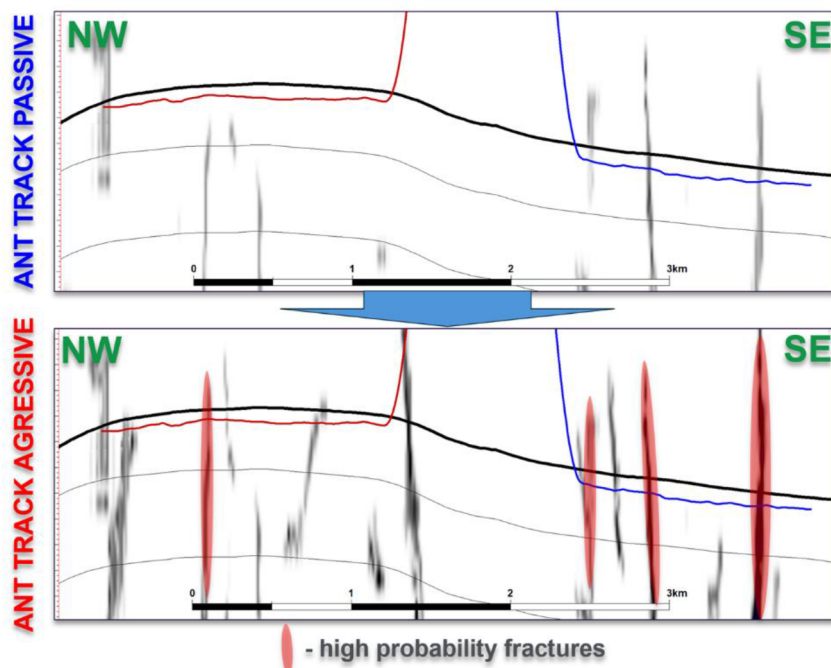


Figure 6—Section view of aggressive and passive Ant-tracks

Seismic Inversion. The main objective of the study was to extract key rock properties—namely lithology, porosity, and total organic carbon (TOC)/kerogen content using the quantitative interpretation workflow. The methodology emphasizes the selection of optimal seismic attributes to enhance reservoir understanding and reduce exploration & appraisal uncertainty.

The initial phase involved comprehensive conditioning of both well log and seismic datasets. Depth-migrated seismic gathers were evaluated for quality and subjected to tailored conditioning procedures. These enhancements were guided by improvements in seismic-to-well ties and the fidelity of derived rock property volumes. AVO (amplitude variation with offset) angle bands were analyzed to determine the optimal range for elastic attribute extraction and thin-bed resolution. This was followed by the generation of preserved, AVO-consistent angle stacks.

Subsequently, a rock physics model was constructed using conditioned well logs to quantify the influence of reservoir quality indicators on elastic responses at the log scale. Multiple seismic reservoir characterization techniques were then applied, including deterministic acoustic and elastic inversions, Bayesian lithology classification, azimuthal inversion for fracture analysis, and pore pressure prediction. Outputs such as Porosity, P and S-impedance and V_p/V_s ratio volumes were derived.

The project was executed in two stages: a feasibility study over a pilot zone containing key wells, followed by full-field implementation. The pilot phase enabled the evaluation and selection of the most effective QI techniques for broader application. Notably, P and S-impedance distributions exhibited strong correlations with porosity, which was validated by the production testing.

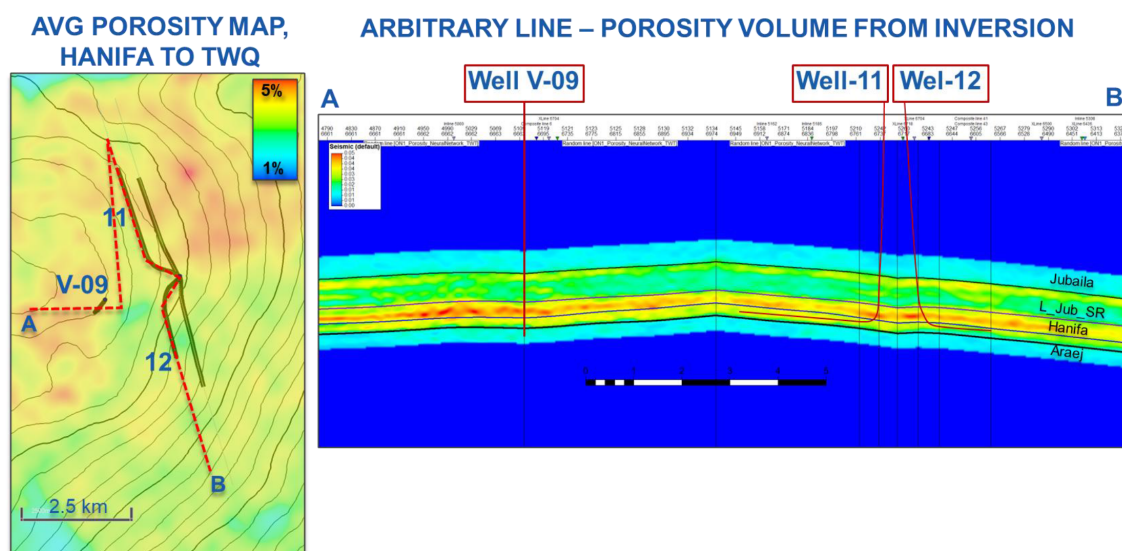


Figure 6a—Hanifa average porosity map derived from inversion overlaid with the structural map and arbitrary line through the vertical and horizontal wells.

Data Integration

To identify sweet spot areas within the unconventional Diyaab gas reservoir, a comprehensive data integration workflow was developed, combining structural, petrophysical, and production datasets. The structural framework was constructed using interpreted horizon maps of the Hanifa, Tuwaiq Mountain, and Araej formations, along with fault and sub-seismic fracture datasets derived from structural attribute volumes such as dip, curvature, and coherence. These attributes were used to delineate zones of enhanced permeability associated with natural fractures. A porosity volume generated from seismic inversion provided spatial distribution of reservoir quality, while well data from 15 producers—including production rates, stimulated lateral lengths, and completion parameters enabled performance benchmarking. Additional analysis was conducted to validate the correlation of the seismic discontinuities and the data received during the frac execution. High grade of correlation was observed between the high confidence fractures (Fig. 6) and the frac stages with the pressure anomalies or casing deformation related with the enhanced permeability intervals.

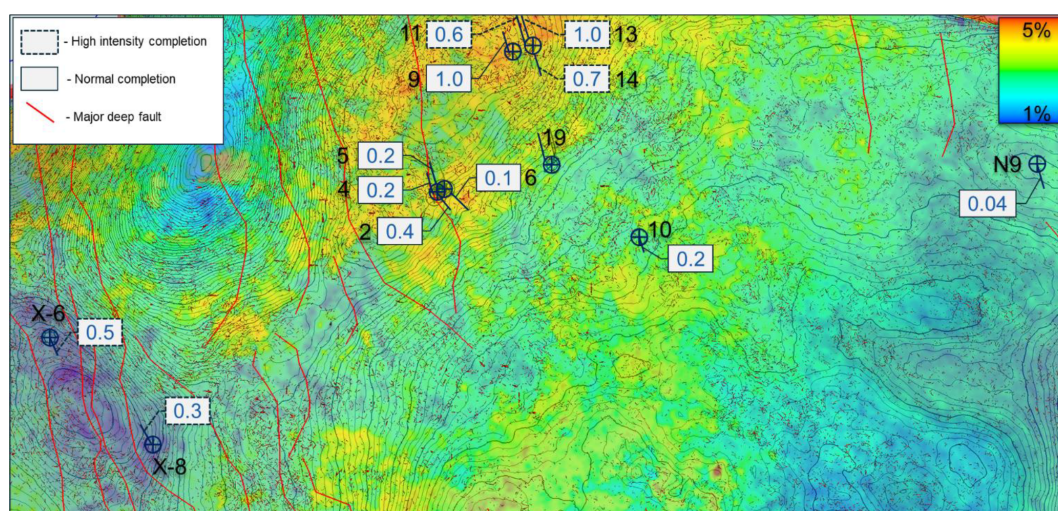


Figure 7—Black contours - Hanifa structural map, color fill – average porosity from seismic inversion, values – normalized gas production

Table 1—Normalized parameters utilized in data analysis

Well	Reservoir	Gas Rate		Porosity	Fractures	Proximity of major fault	Completion intensity	
		Norm WHP	Norm LL				Sand	Water
11	HAN	1.0	0.6	1.0	0.8	Far	0.9	0.8
13	TWQ	0.4	1.0	1.0	0.8	Far	0.8	0.7
14	HAN	0.6	0.7	0.8	0.5	Far	0.9	1.0
19	HAN		0.01	0.8	0.7	Far	0.7	0.7
2	HAN	0.3	0.4	0.9	0.3	Close	0.6	0.9
4	TWQ	0.0	0.2	0.8	0.7	Close	0.5	0.9
5	HAN	0.2	0.2	0.7	0.5	Close	0.5	0.8
6	HAN	0.0	0.1	1.0	0.5	Close	0.5	0.9
9	HAN	0.6	1.0	1.0	0.8	Far	0.3	0.5
10	HAN	0.1	0.2	0.6	0.7	Far	0.3	0.5
X6	HAN		0.5	0.4	1.0	Close	1.0	1.0
X8	HAN	0.0	0.3	0.4	0.5	Close	1.0	1.0
W10	TWQ	0.1	0.2	0.5	0.0	Far	0.4	0.6
N9	TWQ		0.04	0.4	0.5	Far	0.5	0.5
B90	TWQ	0.1	0.0	0.4	0.7	Far	0.5	0.5

Analysis revealed that wells located near major deep faults exhibited poor productivity despite being situated in high porosity zones, suggesting potential gas leakage or water encroachment, which have evidences from the flowback water salinity measurements. Conversely, wells completed with high-intensity designs (>3000 lbs/ft of sand) demonstrated superior performance, even in moderate porosity areas. The most productive wells were those drilled in high porosity zones, away from major faults, with high natural fracture intensity and aggressive completion designs. In contrast, wells in the far eastern part of the study area, characterized by low porosity and diluted Total Organic Carbon (TOC) due to basin slope effects, showed poor performance despite high fracture density—attributed to low Gas Initially In Place (GIIP). These insights were synthesized into a multi-criteria evaluation model, integrating porosity, fracture intensity, fault proximity, completion design, and TOC distribution to delineate optimal drilling targets. This workflow provides a robust framework for sweet spot identification in complex unconventional play.

Conclusions

This study demonstrates the effectiveness of integrating seismic inversion, fracture characterization, and petrophysical analysis to identify high-prospectivity zones within the Diyab unconventional reservoir. By leveraging high-resolution Mega Onshore seismic data calibrated with borehole imaging, advanced mud logging, and production testing, the analysis successfully delineates sweet spots that exhibit enhanced porosity, natural fracture intensity, and favorable geomechanical properties.

The formation evaluation revealed significant heterogeneity across the Diyab Formation, with Hanifa and Tuwaiq Mountain members consistently outperforming Jubaila in terms of porosity*thickness, TOC, and brittleness - key indicators for hydraulic fracturing success. Seismic reservoir characterization, supported by advanced attribute workflows and inversion techniques, provided robust spatial mapping of reservoir quality and fracture networks, even in the absence of visible fault displacement at target depths.

Data integration across structural, petrophysical, and production domains enabled the development of a multi-criteria evaluation model. This model highlights that optimal well performance at the current appraisal stage was achieved in zones with high porosity, intense natural fracturing, minimal fault proximity, and aggressive completion designs. Conversely, wells near major faults or in low-TOC areas showed diminished productivity, underscoring the importance of holistic reservoir understanding.

The methodology outlined in this paper offers a scalable and adaptable framework for unconventional reservoir development, reducing exploration risk and enhancing economic viability. It supports strategic well placement and completion optimization, aligning with the industry's shift toward precision-driven, quality-focused hydrocarbon recovery.

The key outputs of the study:

- It was demonstrated that the gas production is related with the predicted porosity from inversion
- Increase of the natural fracture amount intersecting the lateral has positive correlation with production
- The wells with the low productivity are located in low porosity area, generally have fewer natural fractures
- The best productivity of the wells was obtained in the areas with high porosity, dense natural fracturing and more intensive completion design
- The production of the wells can be affected (pressure interference) due to the proximity to the major basement faults

To consistently validate the impact of the porosity on recovery, which is one of the key metrics of the development phase, long-term production history is required. Once maximum drainage area is seen on the well test data, conclusions can be made regarding the porosity & stimulated rock volume geometry and estimated ultimate recovery (EUR). In current low permeability environment, significant production period is required to establish the well-connected volumes and far zone permeability.

Way Forward

For development of reservoirs exhibiting vertical and lateral heterogeneity such as Jurassic age Diyab agile yet integrated geoengineering workflow is essential. It is required to maximize hydrocarbon recovery while mitigating associated fracturing completion risks such as liner deformation due to occurrence of frac hits or hydraulic fracturing-induced fault activation.

Future workflow would begin with the use of aerial-scale advanced seismic inputs—such as ant-tracking algorithms—to delineate natural fracture networks. When combined with well-scale petrophysical and geomechanical modelling derived data, these inputs would enable the construction of robust 3D static sector models.

Further refinement of these models would integrate drilling-derived parameters, notably Mechanical Specific Energy (MSE), which serves as a proxy for rock confined compressive strength. Additionally, hydraulic fracturing treatment data—such as post-fracture pressure loss and gradient profiles—would be used as critical input for validating model assumptions and enhancing predictive accuracy.

In the subsequent phase, the integration of production data will facilitate the development and validation of dynamic 3D sector models. These models will underpin the optimization of well lateral placement and completion design. This would impact adjustment of fracture staging and frac design to target zones of superior geological quality while avoiding high-risk areas prone to completion damage occurrence along the well trajectories.

To support this initiative, advanced micro-seismic monitoring and high-frequency pressure measurements at the well pad scale are currently undergoing field trial testing. These technologies are expected to provide valuable insights into fracture behavior and reservoir response, further strengthening the model's predictive capabilities.

The described workflow must remain agile and adaptable to align with ADNOC's fast-paced operational tempo, ensuring timely decision-making and continuous optimization throughout the unconventional development lifecycle.

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